

Synopsis

Disease Risk Prediction Using Machine Learning Techniques

Project ID: **PY-GT-48**

Problem ID: **1**

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Repo→

https://github.com/Suryansh-Tayal/Disease_Risk_Prediction.git

<https://github.com/SnehilJoshi007/PBL-AIML>

Problem Description

Machine learning is widely used in healthcare to analyze patient data and assist in early disease prediction. Many diseases such as heart disease can be predicted by studying clinical parameters like age, blood pressure, cholesterol level, blood sugar, heart rate, and other medical attributes. Manual analysis of large patient records is difficult and time-consuming, so an automated system can be very useful.

The objective of this project is to build a simple machine learning classification system that predicts whether a patient is at risk of disease or not using clinical data. The project uses a healthcare dataset and applies preprocessing techniques such as data cleaning and normalization to improve data quality. Three machine learning algorithms—Logistic Regression, K-Nearest Neighbours (KNN), and Support Vector Machine (SVM)—are used for training and testing.

The models are evaluated using accuracy, precision, recall, F1-score, confusion matrix, and ROC curve. This project demonstrates a complete beginner-level machine learning pipeline and shows how AI can support decision-making in the healthcare field.

Steps of Implementation

This project adheres to a straightforward machine learning pipeline. Each phase of implementation is outlined below:

1. Data Collection

A suitable clinical dataset is selected for disease prediction. In this project, the Heart Disease Dataset is used because it is simple, beginner-friendly, and apt for binary classification. It contains patient information such as age, cholesterol, blood pressure, heart rate, and other health parameters.

2. Data Preprocessing

The dataset undergoes cleaning before model training. Missing values are addressed, duplicate records are removed, and data is normalized using scaling techniques. This step enhances data quality and ensures that all features are within a similar range for improved model performance.

3. Model Selection

The following classification algorithms are chosen for their simplicity and ease of understanding for beginners:

- Logistic Regression
- K-Nearest Neighbours (KNN)
- Support Vector Machine (SVM)

4. Training and Validation

The dataset is split into two parts:

- Training Data (80%)
- Testing Data (20%)

5. Models are trained on training data and validated on testing data to evaluate performance on unseen records.

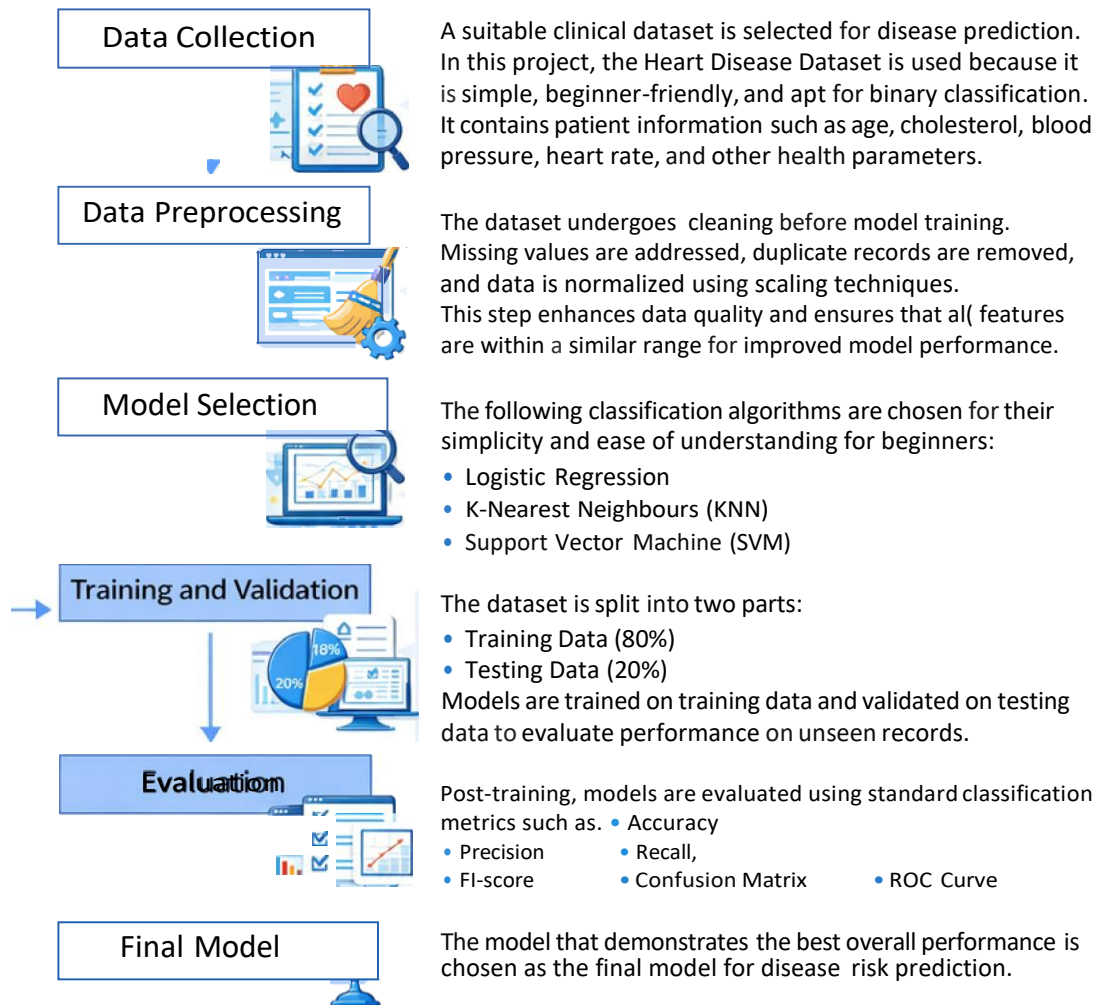
6. Evaluation

Post-training, models are evaluated using standard classification metrics such as:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC Curve

7. Final Model

The model that demonstrates the best overall performance is chosen as the final model for disease risk prediction.



Expected Modules / Libraries to be Used

The following Python libraries and key Scikit-learn modules are anticipated for use in this project:

1. **NumPy**
Utilized for numerical operations and efficient handling of arrays in Python.
2. **Pandas**
Used for loading the dataset from CSV files and for organizing, cleaning, and manipulating tabular data.
3. **Matplotlib**
Employed for plotting graphs such as ROC curves and performance visualizations.
4. **Seaborn**
Primarily used for advanced data visualization, such as drawing the confusion matrix heatmap.
5. **Scikit-learn (sklearn)**
The core machine learning library used for preprocessing, model training, testing, and evaluation.

Important Modules from Scikit-learn

- **train_test_split**: Divides the dataset into training and testing data.
- **StandardScaler**: Normalizes or scales input features to ensure they are within a similar range.
- **LogisticRegression**: Implements the Logistic Regression classifier.
- **KNeighborsClassifier**: Implements the K-Nearest Neighbours (KNN) classifier.
- **SVC**: Implements the Support Vector Machine (SVM) classifier.
- **accuracy_score**: Calculates model accuracy.
- **precision_score**: Computes model precision.
- **recall_score**: Determines model recall.
- **f1_score**: Calculates the model's F1-score.
- **confusion_matrix**: Generates the confusion matrix.
- **roc_curve**: Creates the ROC curve for model evaluation.

Expected Evaluation Metrics and Plots to be Used

The following evaluation metrics and plots will be employed to compare machine learning model performance:

1. **Accuracy**

Measures the correctness of predictions made by the model out of the total predictions.

2. **Precision**

Assesses how many predicted positive cases are actually correct, crucial for reducing false positives.

3. **Recall**

Determines the number of actual positive cases correctly identified by the model, which is vital in medical contexts where missing a disease case can be risky.

4. **F1-Score**

Provides a balanced measure of model performance as the harmonic mean of precision and recall.

5. **Confusion Matrix**

Displays the number of correct and incorrect predictions, aiding in understanding true positives, true negatives, false positives, and false negatives.

6. **ROC Curve**

Used to compare model classification performance at different thresholds.

7. **AUC Score (Optional but Recommended)**

Indicates how well the model can separate classes; a higher AUC suggests better performance.

Plots to be Used

- Confusion Matrix Heatmap
- ROC Curve Plot
- Model Comparison Bar Graph (Accuracy / F1-score)

Required Topics from the Subject

1. Strings

Used for managing text data, handling column names, labels, file paths, and displaying output messages.

2. Pointers

Though primarily a concept in C programming, understanding pointers aids in efficient data handling and memory management.

3. Files

Essential for reading and writing data, as the clinical dataset is stored in a CSV file and loaded into Python for processing.

4. Arrays

Used to store multiple values of the same type, crucial for storing numerical feature values and model input data.

5. Functions

Help divide the program into smaller, reusable blocks. Useful for preprocessing, training models, and evaluating performance.

6. Conditional Statements

Facilitate decision-making in a program, such as checking for missing values or selecting the best model based on accuracy.

Platform Required

The project can be implemented using the following platform:

- **Jupyter Notebook:** Ideal for writing and executing Python code incrementally. It is user-friendly and well-suited for machine learning experiments.

Books and Link Sources

- Scikit-learn Documentation – <https://scikit-learn.org>◆●
- Pandas Documentation – <https://pandas.pydata.org>◆●
- NumPy Documentation – <https://numpy.org>◆
- Matplotlib Documentation – <https://matplotlib.org>◆

Dataset-

Repo-<https://github.com/SnehilJoshi007/PBL-AIML>.

Dataset Overview

Parameter	Details
Purpose	Synthetic dataset for disease risk prediction using patient clinical data. Designed for data cleaning, preprocessing, feature selection, and machine learning classification.
Total Rows	1800
Total Columns	12
Data Quality Design	Includes missing values (~6%) and some noise/outliers to simulate real-world clinical data inconsistencies.

Column Details

Column	Description	Typical Unit	Missing Values (~)
Age	Patient age	Years	~108
Gender	0 = Female, 1 = Male	Binary	~108
BMI	Body Mass Index	kg/m ²	~108
Blood_Pressure	Systolic blood pressure	mmHg	~108
Cholesterol	Serum cholesterol level	mg/dL	~108
Glucose	Blood sugar level	mg/dL	~108
Heart_Rate	Heart beats per minute	bpm	~108
Smoking	Smoking status	Binary	~108
Alcohol_Intake	Weekly alcohol consumption	Units/week	~108
Physical_Activity	Weekly exercise duration	Hours/week	~108
Family_History	Genetic predisposition	Binary	~108
Disease_Risk	Target variable (0 = Low, 1 = High)	Binary	~108

Comprehensive Health Data Analysis Report - Q3 2023											
Section 1: Demographic Overview											
Section 2: Clinical Findings											
Section 3: Laboratory Results											
Section 4: Patient History											
Section 5: Treatment Plan											
Section 6: Follow-up Schedule											
Section 7: Patient Education											
Section 8: Summary and Recommendations											
Section 9: Appendices											
Section 10: References											
Section 11: Glossary											
Section 12: Index											

📌 Quick Summary																						
Column		Min	Max	Comment																		
Age		18	79	Wide age distribution across adults																		
BMI		-15	-40+	Includes underweight to obese cases																		
Blood Pressure		90	180+	Contains hypertension cases																		
Cholesterol		150	300+	Some high-risk values present																		
Glucose		70	190	Includes diabetic range																		
Heart Rate		55	115	Includes abnormal heart rates																		
Physical Activity		0	10	Covers sedentary to active lifestyles																		
Disease Risk		0	1	Balanced classification target																		
1	Age (year)	B	Gender	BMI (kg/m²)	21	Blood_Press (re/mmHg)	C	olesterol (mg/dL)	G	lucose (mg/dL)	H	Heart_Rate (bpm)	S	moking	E	cohol_Intake (units/week)	P	hysical_Activity (hours/week)	F	amily_History	D	isease_Risk
2	34	ma	le	27.9	94/119			277		104		111					4		5/3	no		Low
2	7g	Fem	ale	25.5	106/78			261		104		111					4		6.4	Yes		High
^	29	C	E		161/108			248		103		90	no				1		3.3	unknown		High
5	18	fem	ale	27.9	115/112			191		127		64	N				6		9.1	N		
6	J	mae		35.5	109/117			152		163		79	Yes				6		8.6	N		Medium
7	19	O	thers	19	106-68			160		163		56	No				1		6.7	Yes		High
8	27	F		33.8	139/83			174		170		113	N				4		0	yes		ium
9	18	O	thers	38.3	91-63							SB					8		5.5	H		Low
10	68	Fem	ale	32.4	164/99			258		175		71	No				7		1.3	N		High
11	39	O	thers	11.2	166/62					149		101	N				4		2.7	Yes		High
12	@			38.8	127/101			346		144		78	no				3		9/13	Y		medium
13	32			28.5	159-75			244		124		118	Y				0		3.5	Y		Low
1°	2000	>	es	31.3	104/81			236		103		112					9		8.3	37		Low
15	IBM			32.6	127/92			193		137		96	No				11		5.3	no		Low
18	22	mal+		29.5	107/72			248		89		105	yes				11		9.5	yes		Medlum
17	yg	@	Frs	24.6	100/101			222		191		91	No				6		8.9	NO		Low
^8	35			29.6	151/119			212		73		112	Y				2		9.7	No		Low
19	38	M		33.8	106/118			180		141		70	Yes				2		2.5	yes		Low
20	19	Male		26.8	147/112			228		170		Yes					9		9.3	yes		Low
21	→	o		26.7	162/116			245		88		113	N				5		1	Yes		Medium
••	→	m		18.5	115/73			ysa		ss		113	Y				5		1.2	Y		Medium
23	22	Male		30.3	168/119			311		160		97	yes				4		3	no		High
24	22	Female		17.1	139/66			174		156		55	No				10		8.3	no		High
25	21	M			129/61			273		190		64	no				4		2.7	yes		High
GB	68	M		25.1	96-76			344		122		118	r o				9		2.1	Y		
»	41	r		17.5	109/119			iy2		181		88	u				5		1.7	Yes		Medium
28		male		28.1	143/91			293		193		68	Yes				2		3.5			Medium
29	45	O	thers	26.3	154/84			223		129		79	Yes				2		0.7	no		High
TO	21	ot	S+rs	25.7	175/95			243		83		81	Yes				7		5.2	Y		Low